

Inverse PDE Optimization using Genetic Algorithm

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1 Introduction

Inverse problems for partial differential equations (PDEs) are fundamental in computational physics and engineering, where the objective is to estimate unknown physical parameters from observed data.

In this work, we consider the one-dimensional advection-diffusion equation, which models transport phenomena in fluid dynamics:

$$\frac{\partial u}{\partial t} + v \frac{\partial u}{\partial x} = D \frac{\partial^2 u}{\partial x^2}$$

where $u(x, t)$ is the transported quantity, v is the advection velocity, and D is the diffusion coefficient.

The main objective of this study is to recover the unknown parameters (v, D) from noisy observations using an optimization-based inverse modeling approach.

2 Methodology

2.1 Forward Model

The forward problem is solved using a finite difference scheme with explicit time stepping. Spatial derivatives are approximated using upwind and central difference schemes for advection and diffusion terms respectively.

2.2 Inverse Problem Formulation

Given noisy observations u_{obs} , the goal is to minimize the cost function:

$$J(v, D) = \|u_{sim}(v, D) - u_{obs}\|_2^2$$

2.3 Optimization Method

A genetic algorithm is used to solve the inverse optimization problem. The algorithm evolves a population of candidate solutions through selection, crossover, and mutation to minimize the cost function.

3 Results

The proposed method was evaluated using synthetic data generated with known ground truth parameters:

$$v_{true} = 0.8, \quad D_{true} = 0.05$$

After running multiple stochastic optimization trials, the following results were obtained:

3.1 Parameter Estimation

- Estimated velocity (mean):

$$v_{est} = 0.7804$$

- Estimated diffusion coefficient (mean):

$$D_{est} = 0.0508$$

3.2 Error Analysis

- Mean absolute error in velocity:

$$E_v = 0.0236$$

- Mean absolute error in diffusion coefficient:

$$E_D = 0.0020$$

3.3 Statistical Stability

The standard deviation over multiple runs indicates stable convergence:

- Std of velocity estimates: 0.0266
- Std of diffusion estimates: 0.0013

3.4 Key Observation

The results demonstrate that the proposed genetic algorithm-based inverse solver is able to accurately recover the unknown parameters even in the presence of noise, with low estimation error and stable statistical behavior.

4 Conclusion

This study presented a genetic algorithm-based framework for solving inverse problems in the advection-diffusion equation.

The numerical experiments demonstrate that the proposed method can successfully recover unknown physical parameters with high accuracy:

$$(v, D) \approx (0.7804, 0.0508)$$

compared to the true values:

$$(v, D) = (0.8, 0.05)$$

The low estimation error and stable convergence confirm the effectiveness of combining numerical PDE solvers with evolutionary optimization techniques.

Future work will focus on improving convergence speed and extending the method to higher-dimensional PDE systems.